Beyond Tokenization: Investigating Model Behaviour, Explanation Stability, and Faithfulness under Lexical Perturbations in Devanagari Hindi

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Abstract

Masked language models (MLMs) have become central to multilingual natural language processing (NLP), yet their behaviour and interpretability under non-Latin, morpho-phonetically complex scripts remain critically underexplored. We present a systematic empirical investigation into model behaviour, subword tokenisation dynamics, explanation stability, and explanation faithfulness under controlled lexical perturbations in Devanagari Hindi. Using a triple-annotated benchmark of 200 minimal sentence pairs across three controlled tracks — Valid Word Context ($N = 142$), Synonym Substitution ($N = 38$), and Realistic Spelling Error ($N = 20$) — we evaluate the Indic-dedicated encoder **MuRIL** (`google/muril-base-cased`) and the multilingual generalist **XLM-RoBERTa** (`FacebookAI/xlm-roberta-base`).

Our experiments reveal four primary findings. First, authentic Devanagari spelling errors (matra inversions and ligature splits) induce catastrophic subword fragmentation: single-subword target coverage collapses by -60.0% in MuRIL ($90.0\% \rightarrow 30.0\%$) and by -50.0% in XLM-RoBERTa ($70.0\% \rightarrow 20.0\%$). Second, both models reliably penalise spelling errors under pseudo-log-likelihood (PLL) scoring, but MuRIL exhibits substantially harsher penalties (original preferred in 95.0% of pairs, mean $\Delta\ \mathrm{MeanLogP} = -2.037$) compared to XLM-RoBERTa (65.0%, mean $\Delta = -0.670$). Third, under a corrected visible-target protocol — which resolves an illusory perfect stability artefact ($\rho = 1.000$) induced by target masking — last-layer self-attention achieves $2.1\times$ to $2.7\times$ higher rank stability than Layer Integrated Gradients ($\rho = 0.704$–0.743 vs. 0.263–0.359). Fourth, in discrete context token erasure tests, attention attributions yield superior faithfulness win rates (70.0%–77.5%), whereas Integrated Gradients operates near or below chance (42.5%–60.0%). A correlation analysis confirms that explanation stability does **not** predict explanation faithfulness ($p > 0.05$ across all 8 model–method configurations; $r_s \in [-0.161, +0.270]$).

Keywords: Devanagari Hindi, Masked Language Models, Subword Tokenisation, Pseudo-Log-Likelihood Scoring, Explainable NLP, Explanation Stability, Explanation Faithfulness, Integrated Gradients.